SEE MODEL QUESTION SOLUTIONS

Set 4: Science and Technology
Based on National Examinations Board (NEB) Guidelines

GROUP 'A': Multiple Choice Questions

Q1 (a) Which of the following option shows correct order of classification of phylum of earthworm, frog and butterfly on the basis of their development?

Correct Answer: (iv) Annelida, Arthropoda, Chordata
Explanation: Earthworms belong to Phylum Annelida (less developed), butterflies belong to Phylum Arthropoda, and frogs belong to Phylum Chordata (most developed).

Q1 (b) To identify the blockage of coronary artery which of the following tests is suitable?

Correct Answer: (ii) angiogram
Explanation: An angiogram is a specialized X-ray test specifically used to visualize blockages or narrowing in the coronary arteries.

Q1 (c) Which of the following plants having antimicrobial property is used in the traditional medicine to treat the digestive problem and that related disorders?

Correct Answer: (iii) Calamus (Bojo)
Explanation: Calamus (Rhizome of Bojo) is traditionally renowned for its excellent carminative and antimicrobial properties to treat digestive ailments.

Q1 (d) What is the relation between weight and acceleration due to gravity?

Correct Answer: (iii) $w \propto g$
Explanation: From the formula $w = mg$, since mass (m) remains constant, weight (w) is directly proportional to the acceleration due to gravity (g).

Q1 (e) A ray of light is coming from water to air making an angle of incidence 57° when the critical angle of water is 49°. Which of the following pictures shows the correct route of light?

Correct Answer: (iv) Total Internal Reflection diagram
Explanation: When light passes from a denser medium (water) to a rarer medium (air) and the angle of incidence (57°) is greater than the critical angle (49°), Total Internal Reflection occurs, reflecting light completely back into water at 57°.

Q1 (f) Which type of energy conversion occurs in a generator?

Correct Answer: (i) Mechanical energy into electrical energy.
Explanation: An electric generator converts mechanical work/rotation into electrical energy using electromagnetic induction.

Q1 (g) Under which condition can a flat universe be hypothesized?

Correct Answer: (ii) When the average density is equal to the critical density.
Explanation: According to cosmological models, a flat universe occurs precisely when its actual average mass density equals the critical density value.

Q1 (h) Which one is the general molecular formula of Alkene?

Correct Answer: (ii) C_nH_{2n}
Explanation: Alkenes are unsaturated hydrocarbons containing one carbon-carbon double bond, perfectly corresponding to C_nH_{2n}.

Q1 (i) Study the following statement and arguments about soap:
Statement: Soap is environment friendly.
Argument 1: Soap is biodegradable, hence does not pollute environment.
Argument 2: Soap gives scum in hard water and makes clothes and utensils clean.

Correct Answer: (iii) Statement and argument 1 are correct but argument 2 is wrong.
Explanation: Soap is biodegradable (environment-friendly). However, forming scum with hard water hinders cleaning and wastes soap, making Argument 2 incorrect.

Q1 (j) Which of the following insecticides affects reproductive health?

Correct Answer: (i) BHC (Benzene Hexachloride)
Explanation: Organochlorine insecticides like BHC are persistent organic pollutants that drastically affect hormonal systems and reproductive health.

GROUP 'B': Very Short Answer Type Questions

Q2 (a) Why are variables important in scientific study?

Variables are critical because they help establish verifiable cause-and-effect relationships and maintain standard conditions across experiments.

Q2 (b) Write one difference between analogue signal and digital signal.

An analogue signal is continuous and represents physical measurements, while a digital signal is discrete, carrying data in binary format (0s and 1s).

Q2 (c) Write a difference between the given organisms (DNA and RNA) on the basis of evolution.

RNA evolved first as a self-replicating genetic material (RNA world hypothesis), whereas DNA evolved later as a more stable double-stranded molecular structure for advanced genetic storage.

Q2 (d) Write the conclusion of Hubble's study.

Hubble concluded that the universe is continuously expanding, and distant galaxies are moving away from us at a velocity proportional to their distance.

Q2 (e) Write a difference between drone bee and queen bee on the basis of their work.

The main job of the queen bee is to lay fertile eggs to sustain the colony, whereas the only job of a drone bee is to mate with a virgin queen bee.

Q2 (f) Write a function of worker bee during the age of 4 - 6 days.

During days 4 to 6, a worker bee functions primarily as a nurse bee to feed older larvae with a mixture of honey and pollen.

Q2 (g) What is the problem seen in human health due to high uric acid?

High levels of uric acid lead to crystallization in joints, causing a painful inflammatory condition called gout or gouty arthritis.

Q2 (h) What is the chemical X in the given chemical equation: $H_2SO_4 + 2X \rightarrow Na_2SO_4 + 2H_2O$?

The chemical X is Sodium Hydroxide (NaOH).

Q2 (i) Write a main reason for the use of food preservatives.

Food preservatives are used to inhibit microbial growth and prevent chemical oxidation, thereby extending the shelf life of food products.

GROUP 'C': Short Answer Type Questions

Q3. Test the validity of equation $v^2 - u^2 = 2as$ by unit-wise analysis.

Solution:
LHS units:
Unit of $v^2 = (m/s)^2 = m^2s^{-2}$
Unit of $u^2 = (m/s)^2 = m^2s^{-2}$
Therefore, LHS Unit = m^2s^{-2}

RHS units:
Constant 2 has no unit.
Unit of acceleration (a) = ms^{-2}
Unit of displacement (s) = m
Unit of 2as = $(ms^{-2}) \times m = m^2s^{-2}$

Since the units of LHS and RHS are identical (m^2s^{-2}), the equation is dimensionally valid.

Q4. Write any one similar and a dissimilar character between paddy and mustard.

Similar character: Both paddy and mustard belong to the division Angiosperm as they produce enclosed seeds within fruits.
Dissimilar character: Paddy is a monocotyledonous plant with parallel leaf venation and fibrous roots, while mustard is a dicotyledonous plant with reticulate leaf venation and a taproot system.

Q5. Write differences between DNA and RNA on the basis of function and nitrogen bases.

Feature	DNA (Figure B)	RNA (Figure A)
Nitrogen Bases	Contains Adenine, Thymine, Cytosine, and Guanine.	Contains Adenine, Uracil, Cytosine, and Guanine.
Main Function	Stores and transmits long-term hereditary genetic codes.	Acts directly in protein synthesis and genetic translation.

Q6. Which fertilization process is responsible for surrogate mother to grow the embryo in her womb? How is this process a boon for infertile couples?

Answer: In-Vitro Fertilization (IVF) followed by embryo transfer is the process used. It acts as a boon for infertile couples because it allows couples facing severe physiological reproductive tracks blockages, uterine complications, or low gamete counts to successfully have biological children using a surrogate's healthy womb.

Q7. Suggest any two ways that can be applied to protect endangered animals like one-horned rhinoceros and red panda from habitat loss.

1. Strictly establish, manage, and monitor community-based wildlife corridors and protected conservation sanctuaries.
2. Impose heavy penal consequences on illegal encroachment, deforestation, and poaching activities while promoting sustainable eco-tourism.

Q8. The mass of objects 'A' and 'B' are 2500 kg and 4000 kg respectively. If the distance between their centers is 600 m, calculate the gravitational force exerted between them.

Given Data:

Mass of first object (m1) = 2500 kg

Mass of second object (m2) = 4000 kg

Distance between centers (d) = 600 m

Gravitational constant (G) = 6.67×10^{-11} N m^2 kg^{-2}

Formula:

$F = (G \times m1 \times m2) / d^2$

Step-by-step Calculation:

$F = (6.67 \times 10^{-11} \times 2500 \times 4000) / (600)^2$

$F = (6.67 \times 10^{-11} \times 10000000) / 360000$

$F = (6.67 \times 10^{-4}) / 360000$

$F = 1.852 \times 10^{-9}$ N

Conclusion:

The gravitational force between objects A and B is 1.85×10^{-9} N.

Q9. Where does an object have more weight between Janakpur (lower altitude) and Kathmandu (higher altitude)? Explain with reason.

Answer: An object has more weight in Janakpur than in Kathmandu.
Reason: Weight depends directly on acceleration due to gravity (g), which is inversely proportional to the square of the Earth's radius ($g \mu 1/R^2$). Since Janakpur is at a lower altitude, its surface point is closer to the center of the Earth, resulting in a higher value of g and thus a greater weight.

Q10. In a sample hydraulic machine constructed by using syringes, Syringe A has an area of 0.0005 m^2 and is applied with a force of 20 N. If Syringe B has an area of 0.5 cm^2, find the force produced in piston B.

Given Data:

Area of Piston A (A1) = 0.0005 m^2

Force on Piston A (F1) = 20 N

Area of Piston B (A2) = 0.5 cm^2

Converting A2 to m^2:

A2 = 0.5 / 10000

A2 = 0.00005 m^2

From Pascal's Law:

F1 / A1 = F2 / A2

Step-by-step Calculation:

20 / 0.0005 = F2 / 0.00005

40000 = F2 / 0.00005

F2 = 40000 × 0.00005

F2 = 2 N

Conclusion:

The force produced on piston B is 2 N.

Q11. Complete the given ray diagram for an object placed beyond 2F of a convex lens and write one character of the image formed.

Answer: When an object is placed beyond 2F in front of a convex lens, the light rays parallel to the principal axis pass through the focus (F) on the other side, and rays passing through the optical center pass undeviated. These rays intersect between F and 2F on the opposite side.
Character of image: Real, inverted, and diminished.

Q12. In a transformer, the ratio of number of turns in primary coil and secondary coil is 1:15. If it is connected to a 220 V power supply, calculate the secondary voltage.

Given Data:

Turns ratio (N_p / N_s) = 1 / 15

Primary Voltage (V_p) = 220 V

Formula:

$N_s / N_p = V_s / V_p$

Step-by-step Calculation:

$15 / 1 = V_s / 220$

$V_s = 15 \times 220$

$V_s = 3300$ V

Conclusion:

The secondary voltage obtained from the transformer is 3300 V.

Q13. Write any two differences between alternating current (AC) and direct current (DC).

Alternating Current (AC)	Direct Current (DC)
Magnitude and direction of current change periodically over time.	Magnitude and direction of current remain perfectly constant.
Can be easily stepped up or down using a transformer.	Cannot be stepped up or down via transformers.

Q14. Structural formulas of two hydrocarbons are given (Compound 1: Propyne, Compound 2: Propane). (a) Which one compound is an unsaturated hydrocarbon? (b) Which is a hydrocarbon that can be converted into an alcohol used in thermometers?

(a) Unsaturated hydrocarbon: The first compound containing a triple bond (Propyne / Alkynic chain) is the unsaturated hydrocarbon.
(b) Thermometric match: The compounds can be converted or related to Ethanol. Pure Ethanol is extensively utilized as a safe thermometric liquid in liquid-in-glass thermometers.

Q15. You want to apply electro-refining to get pure silver from impure silver. (a) In which part will you put impure silver between 'A' and 'B'? (b) What is the role of electrolyte 'C'?

(a) Electrode placement: Impure silver is always made the Anode (connected to positive terminal), which is part 'A'. Pure silver sheets are placed at Cathode 'B'.
(b) Role of electrolyte C: Electrolyte C must be a soluble salt of silver (like $AgNO_3$). Its role is to conduct electric ions and transfer pure silver metal from the impure anode onto the pure cathode.

Q16. Equal volume of hydrogen peroxide is kept in two test tubes to produce oxygen. If MnO_2 is added to the first and H_3PO_4 to the second, in which test tube is the reaction faster? Give reason.

Answer: The chemical reaction is significantly faster in the first test tube.
Reason: MnO_2 acts as a positive catalyst which actively lowers activation energy and accelerates decomposition of H_2O_2. Conversely, H_3PO_4 acts as a negative catalyst (inhibitor) that slows down the reaction.

GROUP 'D': Long Answer Type Questions

Q17. What is digital well-being? How do the internet and virtual world affect the physical, social, and emotional health of netizens? Write one point each.

Digital Well-being: It refers to practicing a healthy, balanced, and intentional relationship with digital technology and digital spaces.
1. **Physical Health:** Prolonged screen exposure causes digital eye strain, poor spinal posture, and a sedentary lifestyle.
2. **Social Health:** Excessive virtual substitution leads to reduced face-to-face interaction, dampening real-world social bonds.
3. **Emotional Health:** Constant exposure to toxic cyber interactions or curated highlights causes anxiety and low self-esteem.

Q18. (a) Draw a labelled chart of sex determination. (b) Draw a filial chart crossing between black (BB) and white (bb) rabbits up to the F2 generation and mention the genotypic ratio.

(a) Sex Determination in Humans:
Father (XY) × Mother (XX)
Gametes: Father produces X and Y sperms; Mother produces X and X ova.
Cross Combination Results:
- X (sperm) + X (ovum) → XX (Female Child, 50% probability)
- Y (sperm) + X (ovum) → XY (Male Child, 50% probability)

(b) Monohybrid Rabbit Cross Filial Chart:
Parents: Pure Black (BB) × Pure White (bb)
F1 Generation: All offspring are hybrid black (Bb).
Selfing F1: Bb × Bb
F2 Punnett Square Outcomes:
- BB (Pure Black): 1
- Bb (Hybrid Black): 2
- bb (Pure White): 1
Genotypic Ratio = 1:2:1 (Pure Black : Hybrid Black : Pure White)

Q19. Study the human heart diagram: (a) Name part 'A'. (b) Write a function of part 'B'. (c) Write any two differences between parts 'C' and 'D'.

(a) Part A: Superior Vena Cava / Aorta (depending on specific highlighted vessel context, typically main upper venous inflow).
(b) Function of Part B: Right Ventricle / Left Ventricle wall - pumps deoxygenated blood to the lungs via the pulmonary artery or oxygenated blood to the body organs.
(c) Differences between Atria (C) and Ventricles (D):
- Atria are upper receiving chambers with thin muscular walls.
- Ventricles are lower pumping chambers with highly thick, powerful muscular walls.

Q20. Study the anomalous expansion of water graph: (a) Which property of water does it represent? (b) At what temperature does water have minimum volume? (c) What changes occur to the volume when water at 0°C is warmed to 8°C? (d) Why does water at 4°C overflow if heated or cooled slowly?

(a) Property: Anomalous expansion of water.
(b) Minimum Volume Point: At 4°C, water exhibits its maximum density and absolute minimum volume.
(c) Volume behavior from 0°C to 8°C: When heated from 0°C to 4°C, its volume continuously decreases. Upon further heating from 4°C to 8°C, its volume continuously increases.
(d) Overflow Reason: Since water has its lowest possible volume exactly at 4°C, any change in temperature—whether heating it up or cooling it down—causes its volume to expand, forcing the water to overflow.

Q21. A student faces difficulty reading a book holding it close but reads comfortably when moved away. (a) Identify the defect and its cause. (b) Suggest the lens type and how it helps.

(a) Defect & Cause: The defect is Hypermetropia (Long-sightedness). It is caused by decreased curvature of the eye lens (flat lens) or shortening of the eyeball axis length.
(b) Lens Correction: A Convex lens is used for correction. The convex lens provides converging power to incoming near light rays, ensuring they converge perfectly on the retina instead of focusing behind it.

Q22. Based on the electronic configurations of A, B, C, D: (a) Write the period of element B. (b) Write the chemical nature of element D. (c) Write balanced chemical equation between B and C.

(a) Period of B: B has 3 shells ($1s^2\ 2s^2 2p^6\ 3s^2 3p^1$), so it belongs to Period 3.
(b) Chemical nature of D: D ($1s^2\ 2s^2 2p^6\ 3s^2 3p^5$) has 7 valence electrons, meaning it is a highly reactive non-metal belonging to the halogen group.
(c) Balanced Equation: Element B is Aluminium (valency 3) and C is Chlorine (valency 1).
Chemical Equation: $2Al + 3Cl_2 \rightarrow 2AlCl_3$

Q23. Study the CO_2 lab preparation: (a) Write the formula of compound formed when CO_2 is dissolved in lime water. (b) Write a method to test the gas. (c) Write a use. (d) What happens if delivery tube 'A' is dipped into liquid inside Woulf's bottle?

(a) Chemical Formula: Calcium Carbonate ($CaCO_3$), which creates a milky precipitate.
(b) Laboratory Test: Pass the gas into lime water; it turns milky. Alternatively, introduce a burning matchstick, which extinguishes immediately.
(c) Main Use: Extensively utilized in fire extinguishers and carbonated soft beverages.
(d) Thistle tube context: If the thistle tube / delivery tube end is improperly dipped or sealed, generated gas will escape backwards through the tube instead of collecting in the gas jar.